

HDOC - Day 12 Activity

C Programming Assignment

Menu-Driven Program Using Switch-Case

Problem Statement

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations:

- 1. Find the sum and product of the first and last digits.**
- 2. Count how many digits are lesser than, equal to, and greater than a given digit k.**

Example: For 12345, first + last = $1 + 5 = 6$ and first x last = $1 \times 5 = 5$.

Example: For 12345 and k = 3: lesser = 2, equal = 1, greater = 2.

Algorithm

1. Start.
2. Input a positive integer n.
3. If $n \leq 0$, display an error message and stop.
4. Display the menu with Choice 1 and Choice 2.
5. Input the user's choice.
6. Use switch(choice) to select the required operation.
7. Choice 1: Find last digit using $n \% 10$. Repeatedly divide a copy of n by 10 until the first digit remains.
Calculate sum = first + last and product = first * last.
8. Choice 2: Input digit k. Extract each digit using $\% 10$. Compare it with k

and increase lesser, equal, or greater accordingly. Divide the number by 10 and repeat.

9. For any other choice, display 'Invalid choice'.

10. Stop.

Test Case Table

Test	Input(n)	Choice	k	Expected Output	Result
1	12345	1	-	Sum = 6, Product = 5	Pass
2	9087	1	-	Sum = 16, Product = 63	Pass
3	12345	2	3	Lesser = 2, Equal = 1, Greater = 2	Pass
4	55321	2	5	Lesser = 3, Equal = 2, Greater = 0	Pass
5	12345	3	-	Invalid choice	Pass
6	-123	-	-	Error: Enter a positive integer	Pass

C Program

```
#include <stdio.h>

int main()
{
    int n, choice;
    int temp, first, last;
    int sum, product;
    int k, digit;
    int lesser = 0, equal = 0, greater = 0;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    if (n <= 0)
    {
        printf("Error: Enter a positive integer.\n");
        return 0;
    }
    printf("\nMENU\n");
    printf("1. Find sum and product of first and last digits\n");
    printf("2. Count digits lesser than, equal to, and greater than k\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
```

```
switch (choice)
{
case 1:
last = n % 10;
temp = n;
while (temp >= 10)
temp = temp / 10;
first = temp;
sum = first + last;
product = first * last;
printf("First digit = %d\n", first);
printf("Last digit = %d\n", last);
printf("Sum = %d\n", sum);
printf("Product = %d\n", product);
break;
case 2:
printf("Enter digit k (0-9): ");
scanf("%d", &k);
if (k < 0 || k > 9)
{
printf("Invalid digit k.\n");
break;
}
temp = n;
while (temp > 0)
{
digit = temp % 10;
if (digit < k)
lesser++;
else if (digit == k)
equal++;
else
greater++;
temp = temp / 10;
}
```

```
printf("Count_lesser = %d\n", lesser);
printf("Count_equal = %d\n", equal);
printf("Count_greater = %d\n", greater);
break;
```

default:

```
printf("Invalid choice.\n");
```

```
}
```

```
return 0;
```

```
}
```

Compilation , Execution & Testing

```
Session Actions Edit View Help
(kali@kali)-[~]
$ vi digits.c
(kali@kali)-[~]
$ cc digits.c
(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 12345

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 1
First digit = 1
Last digit = 5
Sum = 6
Product = 5

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 9087

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 1
First digit = 9
Last digit = 7
Sum = 16
Product = 63

(kali@kali)-[~]
$ ./a.out
Enter a positive integer: 12345

MENU
1. Find sum and product of first and last digits
2. Count digits lesser than, equal to, and greater than k
Enter your choice: 2
Enter digit k (0-9): 3
Count_lesser = 2
Count_equal = 1
Count_greater = 2
```

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter a positive integer: 55321  
  
MENU  
1. Find sum and product of first and last digits  
2. Count digits lesser than, equal to, and greater than k  
Enter your choice: 2  
Enter digit k (0-9): 5  
Count_lesser = 3  
Count_equal = 2  
Count_greater = 0
```

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter a positive integer: 12345  
  
MENU  
1. Find sum and product of first and last digits  
2. Count digits lesser than, equal to, and greater than k  
Enter your choice: 3  
Invalid choice.
```

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter a positive integer: -123  
Error: Enter a positive integer.
```